THE SQUARE PRISM AS AN AEROELASTIC NON-LINEAR OSCILLATOR

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SUMMARY

A quasi-steady analysis is made of the transverse galloping of a long prism of square section in a normal steady wind. Experimental stationary aerodynamic forces are used in the theory, which leads to an ordinary differential equation with small non-linearity for the displacement. This is solved by the first approximation method of Krylov and Bogoliubov, and it is found that some characteristics of non-linear oscillators unusual in a mechanical system are predicted, in particular the occurrence of oscillation hysteresis over a range of wind speeds. Quite accurate quantitative agreement between these predictions and wind tunnel measurements on an elastically mounted square prism is described.

1. Introduction

IT is possible to identify several forms of aeroelastic behaviour of bluff cylinders. Best known, perhaps, is the vortex-excited oscillation, in which the cylinder vibrates in a particular mode over a small discrete range of wind speeds containing that at which the fundamental frequency of the Kármán vortex street formed in the wake coincides with the natural frequency of the mode. Almost as well known are the forms generally called galloping, which have in common that the cylinder cross sections are aerodynamically unstable, so that small-amplitude vibrations generate forces which increase the amplitudes to large values. In a continuation of a long term study of bluff cylinder instability (**1, 2**), a particular form of galloping of a cylinder of square section is considered in detail in this paper, both theoretically and experimentally, which exhibits some characteristics of non-linear oscillators unusual in a mechanical system.

The form is pure translation in the direction normal to the plane of the incident wind and the cylinder axis. The square section was chosen because of its known galloping instability (**3**), its fixed separation points, and its symmetry. It is oriented with a side normal to the incident wind. The problem is taken to be two-dimensional, and a considerable effort was made to achieve this condition in the experiments.

2. Notation

y	vibration displacement
t	time
$\dot{y} = \frac{dy}{dt}$	vibration velocity
m	vibrating mass
r	coefficient of viscous system damping
k	spring constant
h	length of side of the square section
l	axial length of prism
ρ	air density
V	air velocity
C_{F_y}	lateral aerodynamic force coefficient
α	angle of attack of section to relative wind
A, B, C, D	coefficients of polynomial approximation to C_{F_y} (tan α)
$Y = \frac{y}{h}$	reduced displacement
$\omega = \sqrt{\frac{k}{m}}$	natural circular frequency
$\tau = \omega t$	reduced time
$\dot{Y} = \frac{dY}{d\tau}$	reduced vibration velocity
$\overline{Y}$	reduced vibration amplitude
ϕ	vibration phase angle
$\beta = \frac{r}{2m\omega}$	reduced damping coefficient
$n = \frac{\rho h^2 l}{2m}$	mass parameter
$\mu = nA$	small parameter in theory of non-linear oscillations
$U = \frac{V}{\omega h}$	reduced air velocity
$U_0 = \frac{2\beta}{nA}$	minimum reduced air velocity for galloping
f	formation frequency of vortices in wake shear layer
$S = \frac{fh}{V}$	Strouhal number
$R = \overline{Y}^2$	dependent variable in non-linear analysis
ψ	polar angle in phase plane
$\mathscr{U}$	designation of unstable focus or limit cycle

$\mathscr{S}$ designation of stable focus or limit cycle

$\epsilon = \dfrac{35nDU}{64}$ parameter in amplitude-time relation

$z = \left(\dfrac{\overline{Y}}{U}\right)^2$ integration variable in amplitude-time relation

p, q real and imaginary parts of complex root of z in integrand

Various subscripts are defined in the text as they occur.

3. Equation of motion

Clearly the aerodynamic excitation of the cylinder must be an asymmetric pressure distribution on the sides of the cylinder afterbody, the part lying in the wake, and it must be recognized that no aerodynamic theory is yet available that could predict such an excitation, given the shape of the section. The most realistic theoretical bluff cylinder flows are the free streamline models, particularly recent ones considered by Roshko (**4**) and Gadd (**5**). These, however, deal only with the flow field outside the wake, and therefore take the pressure to be constant over cylinder afterbodies, thus predicting no lateral excitation.

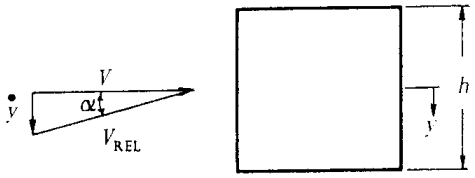


FIG. 1

With no suitable aerodynamic theory available, the oscillation theory uses measured aerodynamic forces in a quasi-steady model. In this model the aerodynamic force in the y-direction on the cylinder moving down with velocity $\dot{y}$ in its vibration cycle in the presence of wind of velocity V (Fig. 1) is taken to be the same as that on the stationary cylinder at angle of attack α in wind of velocity V_{REL}. Assuming a linear elastic system, the differential equation governing the oscillation is

$$m\ddot{y} + r\dot{y} + ky = \tfrac{1}{2}C_{F_y}\rho V^2 hl \tag{1}$$

The aerodynamic force coefficient C_{F_y}, measured on stationary cylinders in the wind tunnel, is a function of α (Fig. 2), and therefore of $\tan\alpha = \dot{y}/V$. C_{F_y} can therefore be approximated by a polynomial in $\dot{y}/V$ over the pertinent range of α. It is an odd function by reasons of symmetry, and requires at least a seventh degree polynomial for a close approximation, because of the five important changes of curvature in the pertinent range $-16° < \alpha < 16°$. Thus we write

$$C_{F_y} = A\left(\frac{\dot{y}}{V}\right) - B\left(\frac{\dot{y}}{V}\right)^3 + C\left(\frac{\dot{y}}{V}\right)^5 - D\left(\frac{\dot{y}}{V}\right)^7 \tag{2}$$

where $A, B, C, D > 0$. Such a polynomial approximation is shown in Fig. 2. If (2) is substituted in (1), and the resulting equation is made dimensionless by dividing through by kh, it becomes

$$\ddot{Y}+Y = nA\left\{\left(U-\frac{2\beta}{nA}\right)\dot{Y}-\left(\frac{B}{AU}\right)\dot{Y}^3+\left(\frac{C}{AU^3}\right)\dot{Y}^5-\left(\frac{D}{AU^5}\right)\dot{Y}^7\right\} \tag{3}$$

The parameters n, equal to half the ratio of displaced air mass to cylinder mass, and β, giving the system damping in the absence of aerodynamic forces, are both very small, of order 10^{-3} in the present system. The

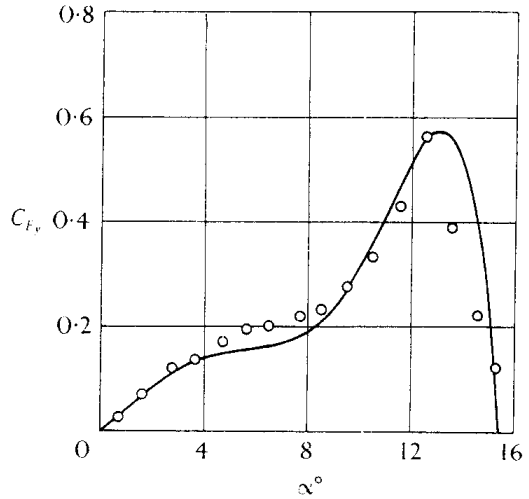


FIG. 2. Lateral force coefficient plotted against angle of attack. ○ Experiment, Reynolds number 22,300 ——— Polynomial Approximation, A = 2·69, B = 168, C = 6,270, D = 59,900

reduced wind speed U is the reciprocal of reduced frequency, and should therefore be fairly large for a quasi-steady theory to be appropriate. In particular, U should be appreciably greater than $1/2\pi S$, where S is the Strouhal number, corresponding to resonance between the system natural frequency and the frequency of vortex formation in the wake of the stationary cylinder. For a square (**6**), S = 0·135, and the corresponding value of U is 1·18.

4. Non-linear analysis

Equation (3) is of the well-known form of an ordinary autonomous differential equation with small non-linear damping

$$\ddot{Y}+Y = \mu f(\dot{Y}), \quad \mu = nA \ll 1 \tag{4}$$

The following method of solution of this equation is essentially that of the first approximation of Krylov and Bogoliubov (see **7**). For $\mu = 0$,

equation (4) is that for the harmonic oscillator, with solution

$$Y = \overline{Y}\cos(\tau+\phi), \quad \dot{Y} = -\overline{Y}\sin(\tau+\phi) \tag{5}$$

where $\overline{Y}$, ϕ are constant amplitude and phase angle. It is assumed that the solution for $\mu \neq 0$, but small, can be obtained as a series expansion in powers of μ

$$Y = \overline{Y}\cos(\tau+\phi)+\mu Y_1(\overline{Y}, \phi, \tau)+\mu^2 Y_2(\overline{Y}, \phi, \tau)+\ldots \tag{6}$$

where $\overline{Y}$, ϕ are now slowly varying functions of τ. The series can be shown to be convergent for sufficiently small values of μ, but in most applications, as here, only the first term is needed to give an accurate description of the oscillation phenomena. In this first approximation method, equations (5) are retained, but $\overline{Y}$, ϕ are considered functions of τ. If equation (4) is multiplied by $\dot{Y}$, the resulting equation can be written

$$\frac{1}{2}\frac{d}{d\tau}(Y^2+\dot{Y}^2) = \mu f(\dot{Y})\,.\,\dot{Y} \tag{7}$$

or, using equations (5)

$$\frac{1}{2}\frac{d\overline{Y}^2}{d\tau} = -\mu f(-\overline{Y}\sin(\tau+\phi))\overline{Y}\sin(\tau+\phi) \tag{8}$$

From equation (8), it is clear that $\overline{Y}$ varies slowly with τ, so that the cycle period is small compared with time intervals during which appreciable changes of amplitude occur. In considering the amplitude-time behaviour, then, it should be satisfactory to average the behaviour over a single cycle, thereby eliminating the harmonics on the right side of equation (8)

$$\frac{1}{2}\frac{d\overline{Y}^2}{d\tau} = -\frac{\mu}{2\pi}\int_0^{2\pi} f(-\overline{Y}\sin\nu)\overline{Y}\sin\nu\, d\nu \tag{9}$$

On inserting the function f from equation (3) and integrating we find

$$\frac{d\overline{Y}^2}{d\tau} = nA\left\{\left(U-\frac{2\beta}{nA}\right)\overline{Y}^2-\frac{3}{4}\left(\frac{B}{AU}\right)\overline{Y}^4+\frac{5}{8}\left(\frac{C}{AU^3}\right)\overline{Y}^6-\frac{35}{64}\left(\frac{D}{AU^5}\right)\overline{Y}^8\right\} \tag{10}$$

The phase angle ϕ can be shown to satisfy an equation similar to equation (9), in the first approximation, namely,

$$\frac{d\phi}{d\tau} = -\frac{\mu}{2\pi\overline{Y}}\int_0^{2\pi} f(-\overline{Y}\sin\nu)\cos\nu\, d\nu \tag{11}$$

Here, the integral vanishes so that ϕ is constant in the first approximation, and may be taken as zero. The behaviour of the square prism as a

non-linear oscillator is therefore determined by equation (10). This can be integrated directly, but it is useful first to examine it qualitatively, both by analysing its form and relating it to behaviour in the $(Y, \dot{Y})$, or phase, plane. The most interesting conclusions to be obtained from such an examination are the conditions of existence and stability of self-excited stationary oscillations.

Now the form of equation (10) is

$$\frac{dR}{d\tau} = aR - bR^2 + cR^3 - dR^4 = F(R) \tag{12}$$

where $R = \overline{Y}^2$, and a, b, c, d depend on the system parameters. The stationary oscillations correspond to $dR/d\tau = 0$ and are therefore given by $R = 0$, and the real positive roots of the cubic

$$a - bR + cR^2 - dR^3 = 0 \tag{13}$$

In the phase plane, $R = 0$, the initial position of equilibrium, is a singular point at the origin known as a focus. The positive roots R_i of equation (13) define trajectories in the phase plane known as limit cycles, here concentric circles of radius $\overline{Y}_{s_i} = \sqrt{R_i}$. If the polar angle in the phase plane is ψ, then

$$\tan \psi = \dot{Y}/Y = -\tan(\tau + \phi) = -\tan \tau \tag{14}$$

so that

$$\frac{d\psi}{d\tau} = -1 \tag{15}$$

and a stationary oscillation of unit circular frequency and amplitude $\overline{Y}_{s_i}$ is represented in the phase plane by a point rotating clockwise with unit angular velocity on a circle of radius $\overline{Y}_{s_i}$.

The stability of equilibrium and of limit cycles is determined by investigating the tendency of the oscillator to return to the original focus or limit cycle after a small displacement δR. The pertinent function is therefore $d\delta R/d\tau$, and from equation (12) this is given by

$$\left.\frac{d\,\delta R}{d\tau}\right|_{R=R_i} = F'(R_i)\,\delta R \tag{16}$$

where the prime indicates differentiation with respect to R. It follows that the sign of $F'(R_i)$ determines the stability of the limit cycle, which is unstable, stable, or neutrally stable according as $F'(R_i)$ is positive, negative, or zero. Here

$$F'(R) = a - 2bR + 3cR^2 - 4dR^3 \tag{17}$$

The stability of the focus at $R = 0$ is determined by the sign of

$$F'(0) = a = nA\left(U - \frac{2\beta}{nA}\right) \tag{18}$$

from equations (17) and (10). If β is negligibly small, the sign is that of A which is positive for the square section, showing it to be unstable at rest at any wind speed in the absence of system damping. Now $A = \left.\frac{dC_{Fy}}{d\alpha}\right|_{\alpha=0}$, and this criterion for instability is equivalent to the one proposed by Glauert (**8**) for auto-rotation of a stalled aerofoil and later by Den Hartog (**9**) for the galloping of electric transmission lines. If $\beta > 0$, the square section is unstable only for

$$U > U_0 = \frac{2\beta}{nA} \tag{19}$$

so that galloping can be eliminated below a given wind speed by sufficiently increasing the system damping.

If equation (19) is satisfied and $a > 0$, then it is clear that there must be at least one limit cycle corresponding to a positive root R_1 of equation (13), since b, c, d are all positive from equations (10) and (2), and the left side of equation (13) will therefore be negative for large R. This limit cycle may be the only one, in which case it is stable, since the slopes of $F(R)$ at successive real roots will alternate in sign, unless a double root intervenes. The phase plane then shows (Fig. 3(a)) an unstable focus encircled by a stable limit cycle, a system with soft self-excitation designated $\mathscr{U}\mathscr{S}$, following Minorsky (**7**), Ch. 7. The corresponding graph of $F(R)$ is also shown in Fig. 3(a). Under these conditions the system oscillates spontaneously from rest with increasing amplitude until a stationary oscillation of amplitude $\overline{Y}_{S_1} = \sqrt{R_1}$ is attained.

Another possibility is that all three roots of equation (13), R_1, R_2, R_3 are real and positive, say $R_3 > R_2 > R_1$. With $a > 0$ ($U > U_0$) this means that R_1 and R_3 represent stable limit cycles separated by an unstable limit cycle, corresponding to R_2. This system is designated $\mathscr{U}\mathscr{S}\mathscr{U}\mathscr{S}$, and the phase plane and graph of $F(R)$ are shown in Fig. 3(b). The transient oscillations, not shown, appear on the phase planes as tightly coiled spirals away from the focus and unstable limit cycle, and towards the stable limit cycles. Inspection of Fig. 3(b) suggests the remaining possible systems. R_2 and R_3 could coalesce to a double root, giving a system designated $\mathscr{U}\mathscr{S}(\mathscr{U}\mathscr{S})$, intermediate between those of Figs. 3(a) and 3(b). Alternatively, R_1 and R_2 could coalesce to a double root, giving a system $\mathscr{U}(\mathscr{S}\mathscr{U})\mathscr{S}$, intermediate between that of Fig. 3(b)

and a final possible system in which R_3 is the only real positive root. The phase plane for this system would be like that of Fig. 3(a).

For a square prism with given mass and damping parameters, n and β, the choice among these possibilities depends on the reduced wind speed U as parameter. This is an example of bifurcation theory in non-linear oscillations. The coefficients a, b, c, d all depend on U, and as it passes

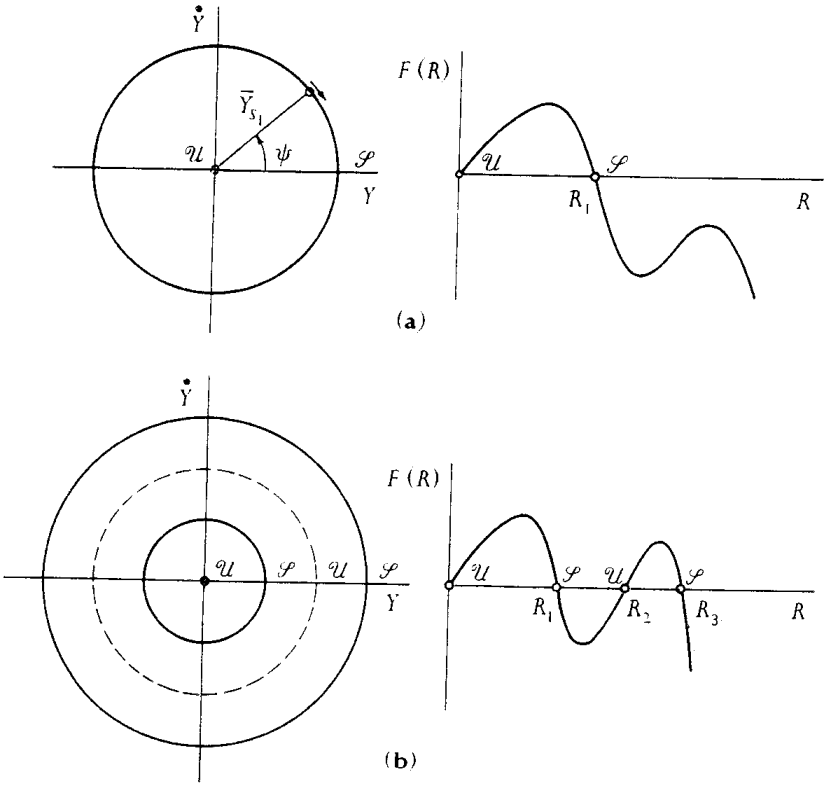


FIG. 3. Phase planes and corresponding functions $F(R)$.

through critical values, the phase portrait of the system changes from one of the forms described above to another.

The dependence of the system on the parameter would commonly be shown on an (R, U) diagram. Here it is convenient to use a $(\bar{Y}_s, U)$ diagram, giving stationary amplitude plotted against wind speed directly (Fig. 4). The figure shows that the system has oscillation hysteresis. At values of $U < U_0$, $\bar{Y}_s = 0$ and the phase portrait is a stable focus. Small oscillations die out and the cylinder remains at rest. At $U = U_0$ the system forks into a form whose phase portrait is like Fig. 3(a), $\mathcal{U}\mathcal{S}$, with the amplitude of the stable limit cycle, $\bar{Y}_{s_1}$, increasing with U, as shown by the curve 012 in Fig. 4. Between U_1 and U_2, however, the phase portrait has become that of Fig. 3(b), $\mathcal{U}\mathcal{S}\mathcal{U}\mathcal{S}$, with

the intermediate forms $\mathscr{U}\mathscr{S}(\mathscr{U}\mathscr{S})$ at U_1, and $\mathscr{U}(\mathscr{S}\mathscr{U})\mathscr{S}$ at U_2. Between U_0 and U_2 the lower limit cycle $\overline{Y}_{s_1}$ is reached from rest. For $U > U_2$ the form becomes $\mathscr{U}\mathscr{S}$, with the single stable limit cycle of amplitude $\overline{Y}_{s_3}$ reached from rest. If the wind speed is then decreased to a value between U_1 and U_2 while the prism is still oscillating, the upper limit cycle is the one reached, until for $U < U_1$ the amplitude drops to that of the lower cycle again. Thus, the path 012343510 containing the hysteresis loop

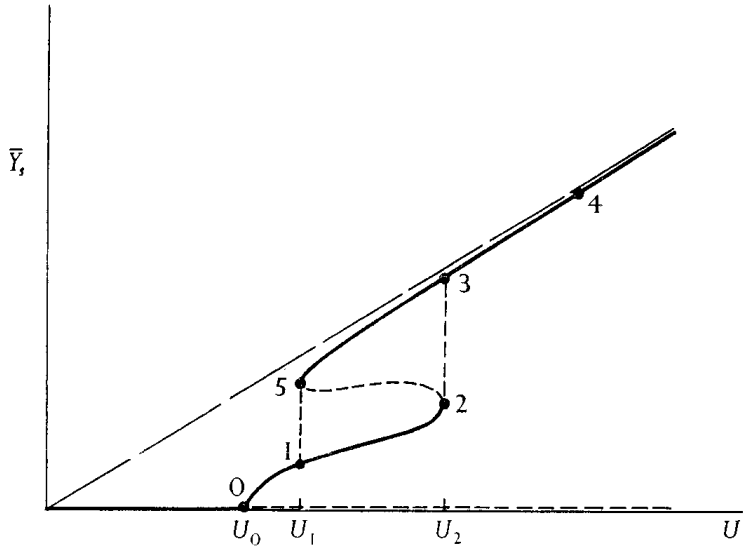


FIG. 4. Amplitude-velocity characteristic showing bifurcations and oscillation hysteresis.

12351 is traversed on the $(\overline{Y}_s, U)$ diagram. The unstable limit cycle in the range $U_1 < U < U_2$ discriminates between the stationary oscillations on the basis of initial amplitude. If the initial amplitude is less than $\overline{Y}_{s_2}$, the stationary amplitude reached is $\overline{Y}_{s_1}$. If it is greater than $\overline{Y}_{s_2}$, the stationary amplitude is $\overline{Y}_{s_3}$.

Two other important characteristics of the $(\overline{Y}_s, U)$ variation can be seen by equating the left side of equation (10) to zero, and dividing through by $nAU\overline{Y}_s^2$

$$\left(1-\frac{2\beta}{nAU}\right)-\left(\frac{3B}{4A}\right)\left(\frac{\overline{Y}_s}{U}\right)^2+\left(\frac{5C}{8A}\right)\left(\frac{\overline{Y}_s}{U}\right)^4-\left(\frac{35D}{64A}\right)\left(\frac{\overline{Y}_s}{U}\right)^6=0 \qquad (20)$$

It is already known that one stationary amplitude $\overline{Y}_{s_3}$ exists for $U > U_2$. Equation (20) shows that as $U \to \infty$, $\overline{Y}_s/U$ becomes independent of U, so that the $(\overline{Y}_s, U)$ curve becomes asymptotic to a line through the origin. Moreover, it can be seen from (20) that a curve of $(nA/2\beta)\overline{Y}_s$ plotted against $(nA/2\beta)U$ would be independent of n and β, so that

experimental values of $\overline{Y}_s$ and U for oscillations of square prisms conducted at various levels of n and β should collapse on a single curve when plotted in this form.

It remains to consider the transient oscillations quantitatively by integrating equation (10). With some rearrangement of terms, it can be integrated to

$$-\epsilon\tau = \int_{z_0}^{z} \frac{dz}{z(z-z_1)(z-z_2)(z-z_3)} \tag{21}$$

where $\epsilon = \dfrac{35nDU}{64}$, $z = \left(\dfrac{\overline{Y}}{U}\right)^2$, $z_0 = \left(\dfrac{\overline{Y}_0}{U}\right)^2$, $z_i = \left(\dfrac{\overline{Y}_{s_i}}{U}\right)^2$, $i = 1, 2, 3$

$\overline{Y}_0$ is the initial amplitude and $\overline{Y}_{s_i}$ are the amplitudes of the limit cycles discussed above. For $U_1 < U < U_2$, the z_i are all real and the integral is

$$-\epsilon\tau = \ln\left\{\left(\frac{z}{z_0}\right)^{\frac{-1}{z_1z_2z_3}}\left(\frac{z-z_1}{z_0-z_1}\right)^{Z_{1,23}}\left(\frac{z-z_2}{z_0-z_2}\right)^{Z_{2,13}}\left(\frac{z-z_3}{z_0-z_3}\right)^{Z_{3,12}}\right\} \tag{22}$$

where $Z_{i,jk} = \dfrac{1}{z_i(z_i-z_j)(z_i-z_k)}$

For $U_0 < U < U_1$, or $U > U_2$, one of the z_i, say z_3, is real and the other two are complex conjugates

$$z_{1,2} = p \pm iq \tag{23}$$

The integral then takes the form

$$-\epsilon\tau = \ln\left\{\left(\frac{z}{z_0}\right)^{\frac{-1}{z_3(p^2+q^2)}}\left(\frac{z-z_3}{z_0-z_3}\right)^{\frac{1}{z_3[(z_3-p)^2+q^2]}}\left(\frac{(z-p)^2+q^2}{(z_0-p)^2+q^2}\right)^{\frac{z_3-2p}{2(p^2+q^2)[(p-z_3)^2+q^2]}}\right\} +$$
$$+\frac{(p^2-z_3p-q^2)}{q(p^2+q^2)[(p-z_3)^2+q^2]}\left\{\tan^{-1}\left(\frac{z-p}{q}\right)-\tan^{-1}\left(\frac{z_0-p}{q}\right)\right\} \tag{24}$$

Equations (22) and (24) give the form of the amplitude-time variation of the prism oscillations, and permit an estimate of the time to build up to a stationary oscillation. This must be defined arbitrarily as, for example, the time to build up from $\overline{Y}_0$ to $0{\cdot}99\overline{Y}_{s_i}$, since the theory gives an asymptotic approach to the focus and limit cycles.

5. Experimental investigation

The appropriate experiments were carried out in the low speed wind tunnel of the mechanical engineering department of the University of British Columbia by the junior author as part of his graduate research program (**6**). The test section was 104 in. long, 36 in. wide, and 27 in. high, and the 1·00 in. square prism tested was mounted vertically at

the centre of the principal cross section. It spanned the height of the test section except for small end clearances at top and bottom.

The static force measurements of Fig. 2 were obtained with the prism supported at the bottom on a conical pivot and mounted at the top on two laterally deflecting leaf springs connected to a turntable. The force was inferred by measuring the displacement of the top of the cylinder with a Daytronic 102–80 differential transformer displacement transducer. The device was carefully calibrated and measurements were made over a range of α of $\pm 20°$, and at four Reynolds numbers, ranging from 22,000 to 36,000. Data were consistent and repeatable, and the only effect of Reynolds number was a small shift of the peak C_{F_y} value to higher α with increasing Reynolds number. Agreement with earlier values (**1**) from integrated pressure distributions on a 2·00 in. square prism was good. The data at the lowest Reynolds number are used in Fig. 2, since the dynamic tests were for still lower Reynolds numbers.

For dynamic tests, the prism was connected by metal tabs piercing narrow transverse slots in the tunnel floor and ceiling to two parallel transverse aluminium shafts mounted in four cylindrical air bearings outside the tunnel. The bearings were mounted on a heavy angle iron frame surrounding the test section, and four light coil springs attached the cylinder and shaft assembly to this frame. The natural frequency of the pure lateral translation to which the system was restrained by the mounting was 6·01 cps, and double amplitudes of 6 in. were possible. The mass parameter n was $4{\cdot}3(10)^{-4}$.

Damping was measured by obtaining decay curves for vibration in still air of the system with the square prism replaced by a streamlined bar of the same mass. Damping was found to be viscous down to the measurable limits of amplitude, and the system value of β was $1{\cdot}07(10)^{-3}$. Additional damping, also viscous in nature, was provided by two electromagnetic eddy current dampers acting on the aluminium shafts.

Stationary amplitudes were measured directly on a scale outside the tunnel past which the model vibrated. Time-amplitude records were obtained using a displacement transducer designed for the purpose. It is essentially an air-core transformer with primary and secondary coils wound on coaxial cylindrical forms with an annular gap between them. The magnetic coupling is varied by inserting the aluminium shaft into the annulus, and the secondary voltage in this way is made proportional to the length of shaft inserted. The output of this displacement transducer was finally recorded on a Minneapolis-Honeywell

Wind speeds were determined from the Betz micromanometer used as the tunnel speed gauge. Reynolds numbers for the dynamic tests ranged from 4,000 to 20,000.

6. Comparison with theory

Measurements of stationary amplitude as a function of wind speed were made at four damping levels. Many measurements were made at each damping level, and the results are plotted in Fig. 5 in the reduced

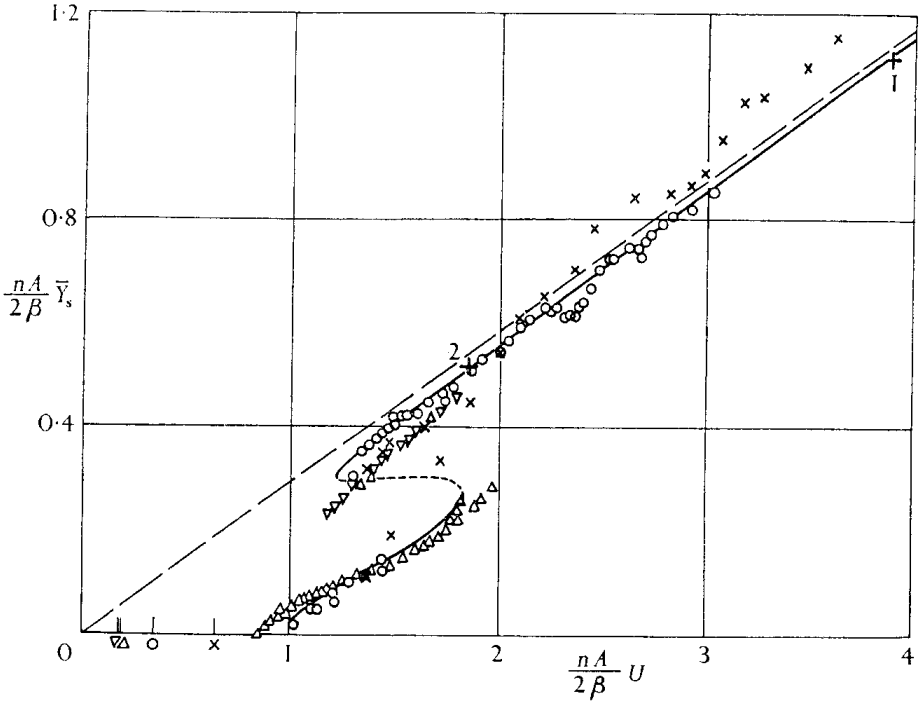


FIG. 5. Collapsed amplitude-velocity characteristic. Theory: ——— Stable limit cycle, – – – Unstable limit cycle. Experiment: × β = ·00107, ○ β = ·00196, Δ β = ·00364, ∇ β = ·00372, +1 β = ·0012, +2 β = ·0032 Reynolds numbers 4,000–20,000.

form described in section 4. Also shown are the theoretical curve and asymptote described in section 4 and shown qualitatively in Fig. 4. The parameters of this curve correspond to the polynomial approximation to the C_{F_y} plotted against α curve shown in Fig. 2. The short vertical marks along the wind speed axis indicate the wind speeds for vortex resonance, $U = 1/2\pi S$, for each damping level (distinguished by the appropriate symbol).

The figure shows that the topological behaviour predicted by the theory of section 4 is observed experimentally. The oscillation hysteresis is present, and the amplitudes and range of wind speeds for the two

stable limit cycles are quite accurately predicted. Moreover, the points for all four levels of damping, in a ratio of about 3·5 : 1, collapse well on to the single theoretical curve. The poorest agreement is for $\beta = 1{\cdot}07(10)^{-3}$, the lowest available value. The data points for this β show a much reduced oscillation hysteresis, and generally worse agreement with the theory. The reasons are that the wind speeds for these tests lie too close to the speed for vortex resonance for the quasi-steady model to be accurate, and that the inaccuracies are magnified in the figure, because the form of plot used magnifies the coordinates of points measured at small β values. Examination of Figs. 2 and 5 reveals that the theoretical curve fits the data slightly better in Fig. 5 than in Fig. 2. If a polynomial is selected to give a closer fit to the data in Fig. 2, the agreement will in fact not be quite so good in Fig. 5 in the region of oscillation hysteresis, but better for large U. The curves used represent a good compromise.

Inspection of the data points along the upper branch of Fig. 5 shows a small consistent variation about the theoretical curve, suggesting resonance effects not accounted for in the quasi-steady model. Since the curve approaches the asymptote closely at higher wind speeds, a convenient approximate formula for estimating the galloping amplitude of a lightly damped square cylinder is seen to be

$$Y_s \simeq 0{\cdot}3U, \qquad \text{or} \qquad \bar{y}_s \simeq 0{\cdot}3\,\frac{V}{\omega}\ \text{ft} \tag{25}$$

where V is in ft/sec., and ω is in radians/sec.

Fig. 6 shows two comparisons of theoretical and experimental amplitude-time curves. The experimental curves were obtained directly from the Visicorder traces, and the theoretical curves were calculated from equation (24). The first comparison is made for a wind speed well removed from the region of oscillation hysteresis, point 1 in Fig. 5.

The theoretical and experimental curves are matched at an initial amplitude $\overline{Y}_0 = 0{\cdot}05$, since this seemed characteristic of the start of galloping when the prism was released by turning on the air supply to the air bearings. The two curves are in good agreement, except that the theoretical curve has a slight hump at about 150 cycles, not observed on the experimental trace. This is a residual effect of the lower limit cycle that would occur at a lower wind speed, and its absence from the experimental trace is probably due to the low damping level of the test, $\beta = 1{\cdot}2(10)^{-3}$; this again brings the wind speed range in which oscillation hysteresis would otherwise occur at that β too close to the speed for vortex resonance. The large number of cycles required for the oscillation

to become stationary, about 300, justifies the assumptions of the non-linear analysis.

The other comparison is made for a wind speed just beyond the range of oscillation hysteresis, point 2 in Fig. 5. The parameters of the theoretical curve used here are slightly different from the ones used previously. Because of the sensitivity of conditions in this region, they were adjusted so that the stationary amplitudes agreed, and the experimental wind

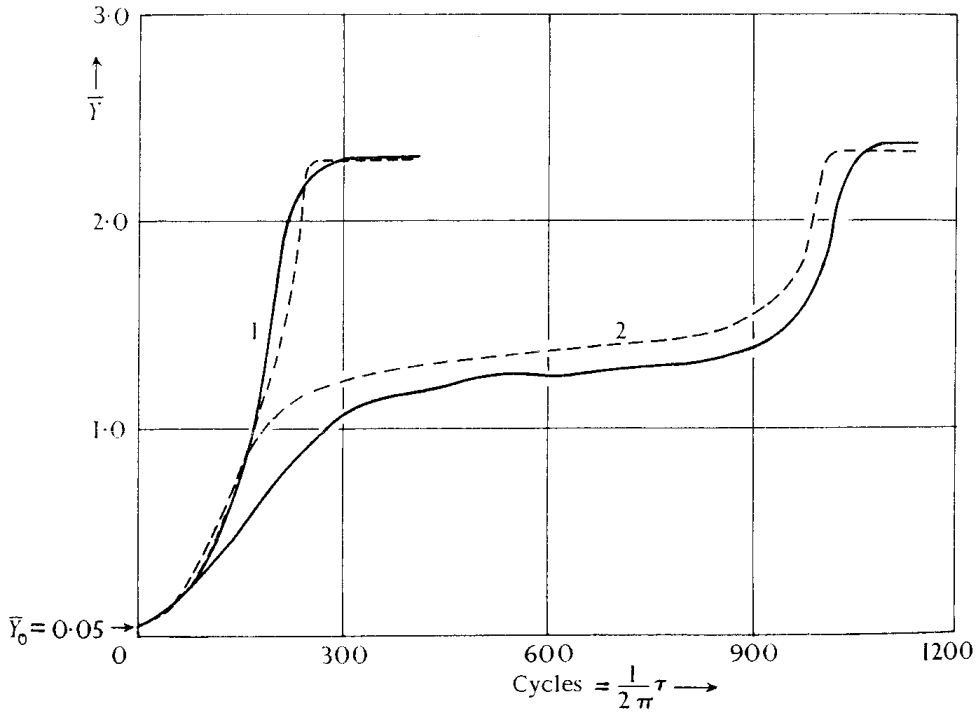


FIG. 6. Amplitude-time characteristics. Theory – – – Experiment ———.

speed lay just beyond the knee of the theoretical amplitude vs wind speed curve. Point 2 in Fig. 5 is shown in the position it would occupy if it, rather than the theoretical parameters, had been adjusted. The theoretical curve is seen to agree well with the observed trace, quite accurately showing the plateau arising from the residual effect of the lower limit cycle, and giving a good estimate of the build-up time, again very long in terms of cycles.

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